Malocclusion and Jaw Joint Disorders: A Look into Auto-segmentation and Quantification of Medical Scans

1st David Jr Sim
Department of Computer Science
University of California, Santa Barbara
Santa Barbara, United States
0009-0002-0073-1447

***Abstract*—Many tasks in dentistry require the segmentation of medial images. Human-provided segmentation has proven to be slow, prone to errors, inconsistent, and require specialized training. The overarching purpose of this study is to utilize 3D mouth scans with jaw disk MRI images to determine if there is a significant relationship between malocclusion (ie. the misalignment of the upper and lower teeth) and temporomandibular joint disorders. However, as equally important in this study is the utilization of auto-segmentation techniques on both 3D point clouds and MRI images to achieve this analysis. For this, the architecture called TSegNet was utilized as a starting point. Afterwards, post processing includes robust regression to quantify the misalignment of these teeth points. Due to unforeseen circumstances during the research, as well as time constraints, the MRI section of this research is still ongoing as of writing this paper. However, the idea is to utilize a Unet architecture, trained on 512×512 *.png*'s, gray-scaled and normalized. The mask of the articular disk segmented will then be used to determine the location of the articular disk and its relative position against the articular fossa to quantify misalignment. Results from TSegNet indicate that the model does phenomenally on standard teeth and their alignment. However, the model falls short when coming into contact with heavily misaligned datapoints.**

***Index Terms*—articular disk displacement, dental, dental scan, machine learning, malocclusion, medical image analysis, MRI, quantification, segmentation, teeth segmentation**

I. Introduction

3D mouth scans are a common tool for dentists to diagnose and treat a variety of symptoms [1]. Many of these tasks requires the segmentation of individual teeth. Similarly, MRI imaging is used in addition when the entire structure of the jaw is concerned [2]. These images also requires segmentation of important and relevant body parts. In both cases, manual segmentation across the medical fields has proven to be slow, prone to errors, inconsistent, and require specialized training [3]. With the rise of artificial intelligence and machine models, there has been a significant push in recent years to adopt more automated solutions [4].

The overarching purpose of this study is to utilize 3D mouth scans with jaw disk MRI images to determine if there is a significant relationship between malocclusion (ie. the misalignment of the upper and lower teeth) and temporomandibular joint disorders (specifically the displacement of the articular disk) [5]. However, as equally important in this study is the utilization of auto-segmentation techniques on both 3D point clouds and MRI images to achieve this analysis.

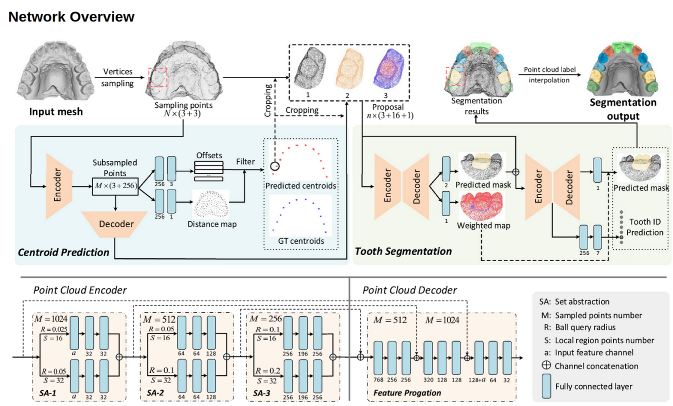

Fig. 1. The network diagram of TSegNet as presented in [6].

As of writing this paper, the study is not yet complete. Thus, no conclusions can be drawn as to the purpose of this study. However, this paper will focus more on the technical aspect within the study. Specially, an analysis of the different techniques used to automatically preform segmentation on 3D point cloud scans of patients' mouths as well as on MRI images of patients' temporomandibular joint will be conducted and different results utilizing these methods will be presented.

II. Methods

A. Teeth Segmentation

The first subtask requires the segmentation of each individual teeth to analyze their position, alignment, and relative displacement. For this, the architecture called **TSegNet** from [6], shown in 1, was utilized as a starting point. Due to the rushed timeline, pretrained weights were implemented as a proof-of-concept step. Retraining of the model utilizing custom data that is more relevant to this current study will be necessary in the future to provide high efficacy results. This retraining will be completed at a later date.

As seen in 1, this network consists of two distinct parts: centeroid analysis and teeth segmentation. While other, generalized point cloud segmentation models [7] [8] [9] [10] have a direct approach to segmenting an unknown domain of images, **TSegNet** is specifically curated for mouth and teeth segmentation. In the first part, it preforms an analysis of the scan and gives a estimated prediction of where the center of mass (centeroid) of each tooth could be. This estimation is then used to make an educated crop of the point cloud to isolate the tooth and to create a heat map of where the tooth could be. These extra data points are then fed into the actual tooth segmentation model to be processed. It requires the *.obj* file format due to the format's easy manipulation and extraction of 3D data. The model also requires the orientation of the scan to be in a specific order. The model also has its own Docker container base that can ease the setup process.

Afterwards, these segmentation points are used to recalculate the centeroids of each tooth, confident that these points are correct, and apply robust regression onto these points to create a jaw curve. This study uses three types of robust regressor: huber [11], ransac [12], and none. Robust regressors are used to fit the predicted jaw-shape curve due to the necessity of filtering out misaligned teeth. This curve will then be compared against to determine misalignment of each individual tooth. Previous literature suggest that a high degree polynomial is sufficient in estimating the shape of the human jaw. As such, from the suggestions of [13], the following polynomial was chosen, with a and b as hyperparameters.

$$\hat{y} = ax^6 + bx^2 \tag{1}$$

This allows for easy comparison and, given that the equipment used to take the teeth scans already align the two jaw halves relatively, it is trivial to quantify the displacement of the two jaw arches at this point.

B. Jaw MRI Segmentation

Due to unforeseen circumstances during the research, as well as time constraints, the MRI section of this research was not completed. However, the approach will still be discussed here.

Utilizing images of MRI scans, a u-net architecture [14] will be will be trained on 512×512 *.png* gray-scale, normalized preprocessed versions of these scans with 32 input features for its first layer. The rest will follow the specified flow in [14]. Training will be conducted over 1000 epochs with batches of 32 on a novel dataset gathered from real patients.

The expected output will be a mask of the same size with the articular disk segmented out from the image. While this approach gives a lot of empty space in the mask, there is no theoretical worry that the model may have a hard time optimising. However, if this is found practically, then the segmentation task will also include identifying the articular fossa, the articular eminence, and the mandibular condyle to give the model more context to work with.

The mask will then be used in the post processing to determine the location of the articular disk and its relative position against the articular fossa. This will reveal if the disk is normal or suffering from a temporomandibular disc disorder. This displacement will be quantified and utilized throughout the overarching research purpose.

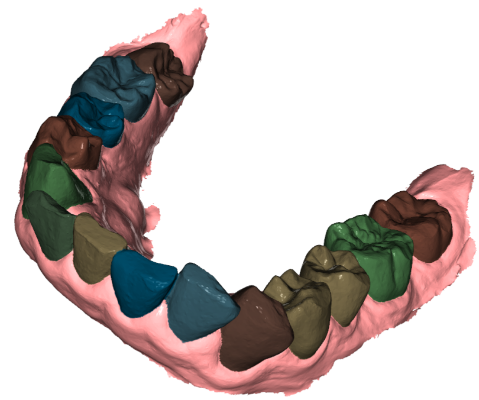

Fig. 2. A good result from the model. All teeth are segmented correctly.

III. Results

A. TSegNet

Utilizing the model was very simple. If the orientation of the scan is guaranteed, then, in most cases, the model preforms well. Throughout the dataset that was used, there was no efficient way to guarantee consistency in orientation. As such, some of the bad results may stem from an orientation issue due to the model's sensitivity to this hyper parameter.

Figures 2, 3, 4, and 5 all show examples of good results from the model. The model does phenomenally on standard teeth and alignment. This is likely due to the fact that the pretrained model was created on a pubic dataset of standard teeth with no abnormalities. As seen in 3, while the alignment of the teeth are important, there is still some robustness present within the model to work around misalignment to a certain extent. In 5, there are some missing data points within the point cloud. This model is robust enough to work around this issue, given that the missing data points are not major as in this figure.

Outside of the standard teeth and their alignment, the model struggles significantly, as seen in 6, 7, 8, and 9. In 6, the misalignment is too great for the model to handle. As mentioned previously, the model was trained on standard alignment of teeth. As such, this is a situation not covered within its domain, causing it to fail to recognize the protruding and overlapping front teeth. In 7, due to the irregular shape of the jaw, the model failed to see and segment the back molar teeth. In 8, the crown cap could not be segmented due to it not being a tooth. The model has not been trained to include foreign objects within its segmentation predictions. When it sees a foreign object, it will treat it as if it is apart of the

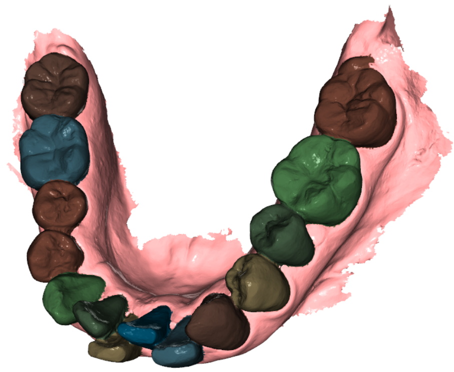

Fig. 3. A good result from the model. Even with some misalignment, the model is robust enough to detect it and segment the teeth properly.

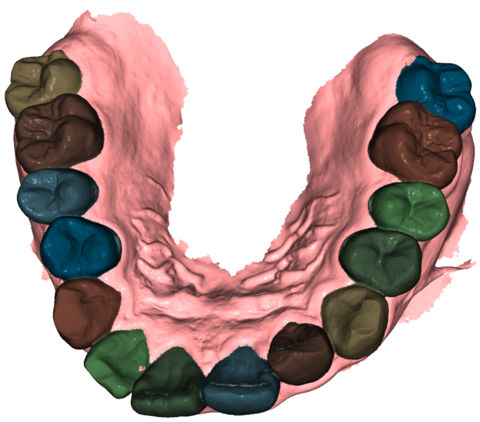

Fig. 4. A good result from the model. Another example of the model and its ability to successfully segment standard teeth.

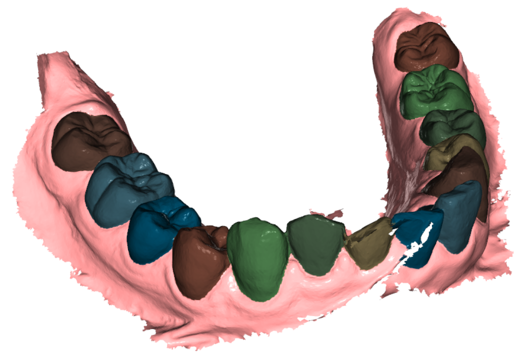

Fig. 5. A good result from the model. Even with some missing points in the scan, the model is still able to work around this defect and segment properly.

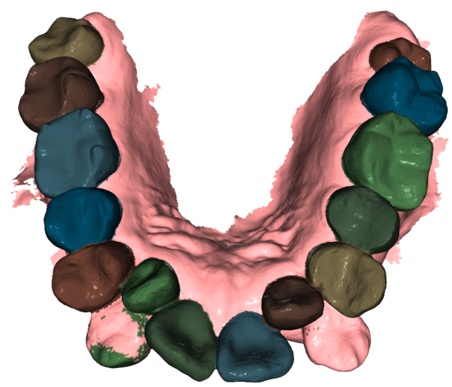

Fig. 6. A bad result from the model. The misalignment of this scan is too great for the model to handle.

teeth and continue to segment as best as it can. This could be mitigated with a broader dataset that have clearly defined ground truth labels that describe how to handle ambiguous situations such as this. As for 9, the messiness of the scan confused the model, showing the importance of removing any irrelevant data from the model.

With the subset of data that contained the good results, the post processing steps are applied. The standard jaws, such as 10, were no problem to for any of the three regresses to handle. Each had a great attempt and all were visually acceptable. The predictions from both huber and ransac were slightly better than none, with no regressor showing a slight jaw skew to the left. On the other hand, the jaws with high misalignment were tougher for the the regressors to handle, such as in 11. For situations such as this, it is surprising to see that no robust regression is better than any robust regression. There might be another method more suitable for this tasks, however, ransac and huber are not those methods. Currently, these results show that a more robust regressor is needed to predict the shape of those with a bad alignment. In the worst case, a new model will need to be trained for this task specifically.

B. U-net

As mentioned before, due to extraneous circumstances, as of writing this report, the U-net [14] has not be trained to a satisfactory level to be presented in this paper. Thus, no results can be presented in this section.

IV. Conclusion

While the field of segmentation using computer vision has improved dramatically within recent years, results show that there are still a lot of validity in results that need to be

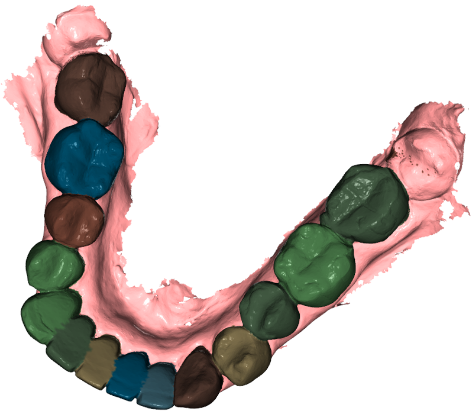

Fig. 7. A bad result from the model. Due to the shape of the jaw, the model was unable to detect and segment the molars.

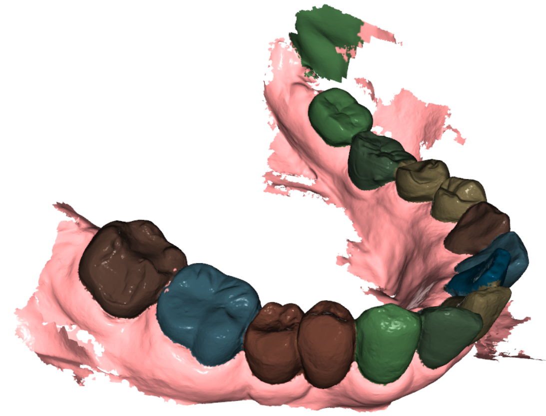

Fig. 9. A bad result from the model. The weird and messy scan confused the model.

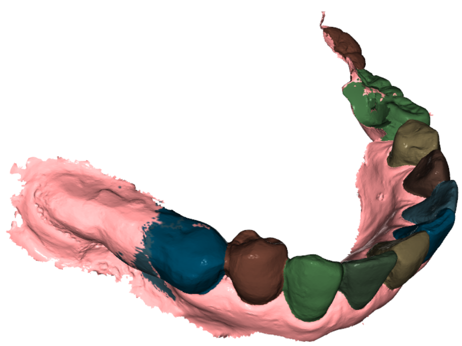

Fig. 8. A bad result from the model. The crown cap is a major obstacle for the model.

improved before clinical trials can happen. 3D segmentation of teeth scans can be an indispensable tool for dentists and medical professionals in saving time and resources while improving accuracy and workflow. Similarly, segmentation of MRI images can support technicians and professionals in their diagnosis. Although this paper only scratches the surface of current improvements in this field, the future of implementing some of the workflows shown here may not be too far off.

References

[1] M. Javaid, A. Haleem, and L. Kumar, "Current status and applications of 3D scanning in Dentistry," Clinical Epidemiology and Global Health, vol. 7, no. 2, pp. 228–233, Jun. 2019. doi:10.1016/j.cegh.2018.07.005

[2] R. Reda et al., "An update of the possible applications of magnetic resonance imaging (MRI) in Dentistry: A literature review," Journal of Imaging, vol. 7, no. 5, p. 75, Apr. 2021. doi:10.3390/jimaging7050075

[3] T. N. Andersen et al., "Accuracy and precision of manual segmentation of the maxillary sinus in Mr Images—a method study," The British Journal of Radiology, vol. 91, no. 1085, Mar. 2018. doi:10.1259/bjr.20170663

[4] J. Im et al., "Accuracy and efficiency of automatic tooth segmentation in digital dental models using Deep Learning," Scientific Reports, vol. 12, no. 1, Jun. 2022. doi:10.1038/s41598-022-13595-2

[5] R. Takaoka et al., "Relative risk of positional and dynamic temporomandibular disc abnormality for osteoarthritis—magnetic resonance imaging study," Journal of Oral Rehabilitation, vol. 48, no. 4, pp. 375–383, Jan. 2021. doi:10.1111/joor.13138

[6] Z. Cui et al., "TSegNet: An efficient and accurate tooth segmentation network on 3D dental model," Medical Image Analysis, vol. 69, p. 101949, Apr. 2021. doi:10.1016/j.media.2020.101949

[7] R. Q. Charles, H. Su, M. Kaichun, and L. J. Guibas, "PointNet: Deep Learning on point sets for 3D classification and segmentation," 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Jul. 2017. doi:10.1109/cvpr.2017.16

[8] X. Hu and X. Xie, PointNet++ network with contextual feature and mutual learning for point sets, Mar. 2024. doi:10.20944/preprints202403.0743.v1

[9] Y. Wang et al., "Dynamic graph CNN for learning on point clouds," ACM Transactions on Graphics, vol. 38, no. 5, pp. 1–12, Oct. 2019. doi:10.1145/3326362

[10] H. Zhao, V. Koltun, P. Torr, J. Jia, and L. Jiang, Point Transformer, 2021.

[11] E. M. Ronchetti and P. J. Huber, Robust Statistics. Hoboken, N.J: Wiley, 2013.

[12] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," Readings in Computer Vision, pp. 726–740, 1987. doi:10.1016/b978-0-08-051581-6.50070-2

[13] H. Noroozi, T. H. Nik, and R. Saeeda, "The Dental Arch Form Revisited," Angle Orthod, vol. 71, no. 5, pp. 386–389, 2001. doi: https://doi.org/10.1043/0003-3219(2001)071<0386:TDAFR>2.0.CO;2

[14] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," Lecture Notes in Computer Science, pp. 234–241, 2015. doi:10.1007/978-3-319-24574-4_28

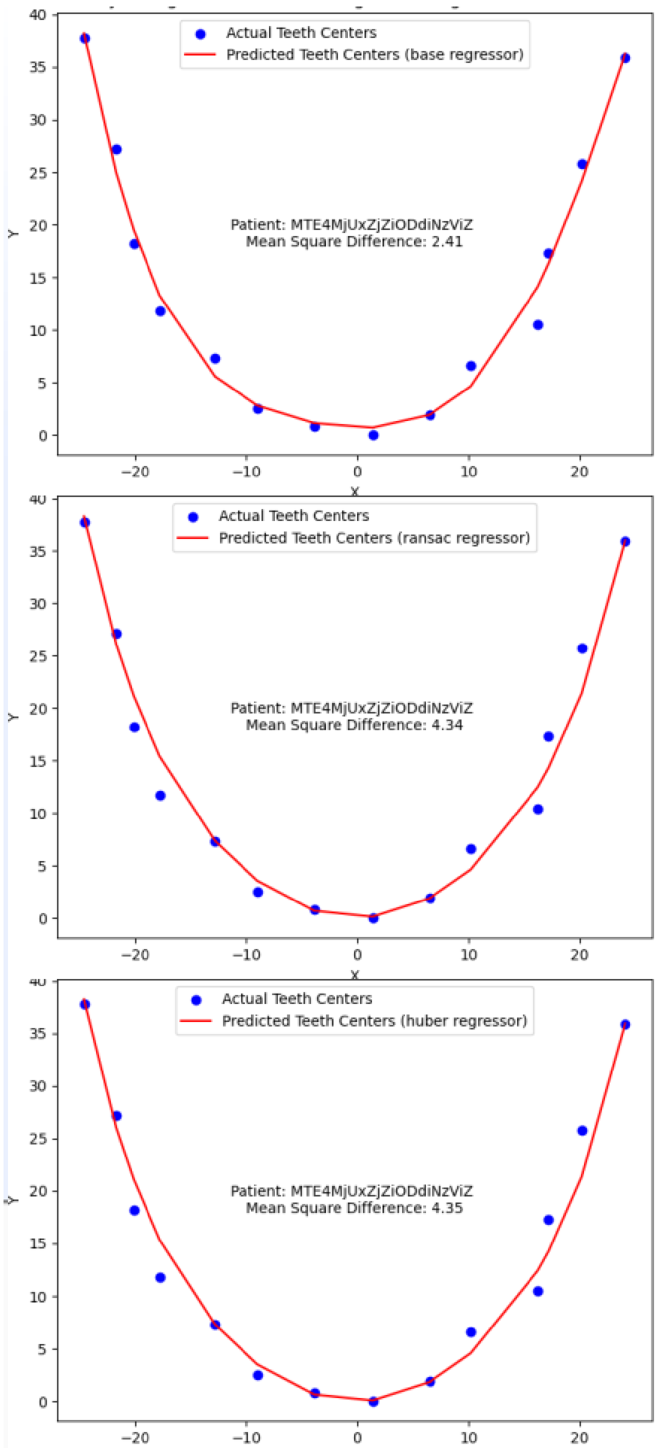

Fig. 10. Good regression and predicted jaw results from the post processing step.

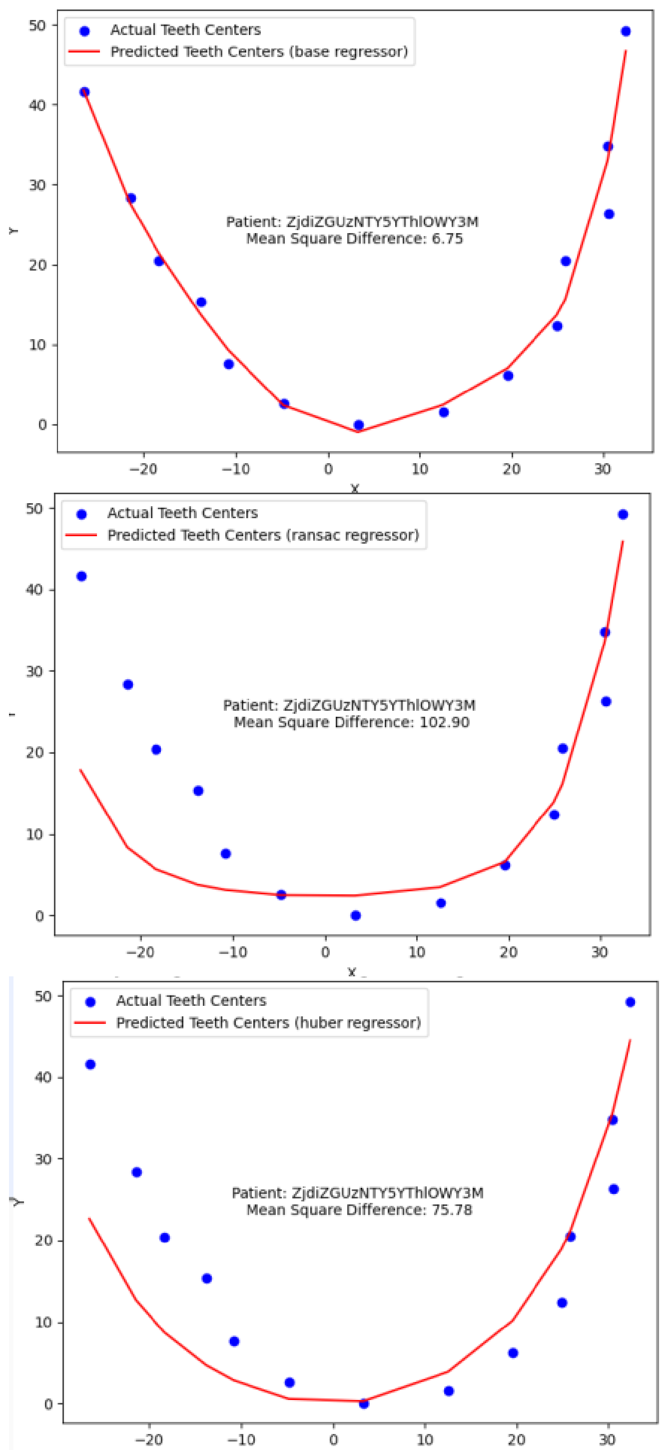

Fig. 11. Bad regression and predicted jaw results from the post processing step.